Multivariable Function

↳ Real-valued function of more than one input variable.

$f(x_1, \dots, x_n) = y$ ← output variable

↑ independent parameters

Any situation i.e. a "number" you could measure that changes based on multiple factors.

Function must always return the same value for the exact same inputs.

The parameters determine the value

↳ ① What parameters are necessary

↳ ② How does changing each variable change the final value.

$f(x_1, x_2, x_3)$

$$f(1, -3, 500) = 10$$

$$f(1.1, -3, 500) = 10.25$$

$$f(1, -3.5, 500) = 11$$

$$f(1, -3, 550) = 8$$

$$\frac{\Delta f}{\Delta x_1} = \frac{10.25 - 10}{1.1 - 1} = \frac{0.25}{0.1} = 2.5 \approx \left.\frac{\partial f}{\partial x_1}\right|_{(1,-3,500)} = 2.5$$

$$\frac{\Delta f}{\Delta x_2} = \frac{11 - 10}{(-3.5) - (-3)} = \frac{1}{-0.5} = -2 \approx \left.\frac{\partial f}{\partial x_2}\right|_{(1,-3,500)} = -2$$

$$\frac{\Delta f}{\Delta x_3} = \frac{-2}{50} = -0.04$$

$$\left.\frac{\partial f}{\partial x_3}\right|_{(1,-3,500)} = -0.04$$

$$f(2, -4, 400) \approx f(1, -3, 500) + \frac{\partial f}{\partial x_1}\cdot(2-1) + \frac{\partial f}{\partial x_2}(-4-(-3)) + \frac{\partial f}{\partial x_3}(400-500)$$

$$= 10 + 2.5(1) + -2(-1) + (-0.04)(-100)$$

$$= 10 + 2.5 + 2 + 4$$

$$= 18.5$$

first partials only say how the output changes from one specific input.

We need higher derivatives (≥ 2) to understand how one parameter affects the other parameters.

$$\nabla f = \left\langle \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right\rangle, \quad \Delta x_i = \langle x_1 - a_1, \dots, x_n - a_n \rangle$$

$$f(x_1, \dots, x_n) \approx f(a_1, \dots, a_n) + \nabla f \cdot \Delta x_n$$

$$T_a(x_1, \dots, x_n) = f(a_1, \dots, a_n) + A\sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(x_i - a_i) + \sum_{i=1}^{n}\sum_{j=1}^{n} \frac{\partial^2 f}{\partial x_i \partial x_j}(x_i - a_i)(x_j - a_j) + \cdots$$

$$f(x_1, \dots, x_n) \rightarrow \frac{\partial f}{\partial x_1}$$ = "Derivative but pretend all the variables but one are constants"

$$\frac{\partial f}{\partial x_2}(x_1, \dots, x_n)$$

$$\vdots$$

$$\frac{\partial f}{\partial x_n}$$

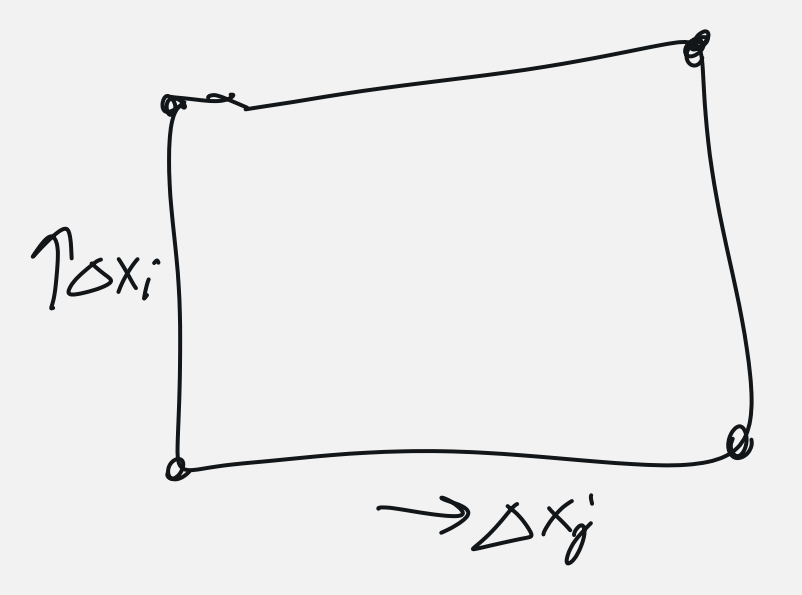
1 function n 1st order partials   Take all 1st order partials → n^2 2nd order partials

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \underset{\overset{\uparrow}{j}}{\frac{\partial}{\partial x_i}}\left(\frac{\partial f}{\partial x_j}\right)$$

(H) Hessian Matrix $n \times n$

$$= \begin{array}{cc} i \rightarrow & \begin{bmatrix} & & \\ & \boxed{\frac{\partial^2 f}{\partial x_i \partial x_j}} & \\ & & \end{bmatrix} \end{array}$$

If all $\frac{\partial^2 f}{\partial x_i \partial x_j}$ are continuous on domain of interest, then $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$



H is symmetric then. (This usually happens for real-world stuff)

which means by defintion of symetric that it is equal to its transpose

~~If~~ H is "positive-semidefinite" then f is convex   cirtical point + convex = minimum

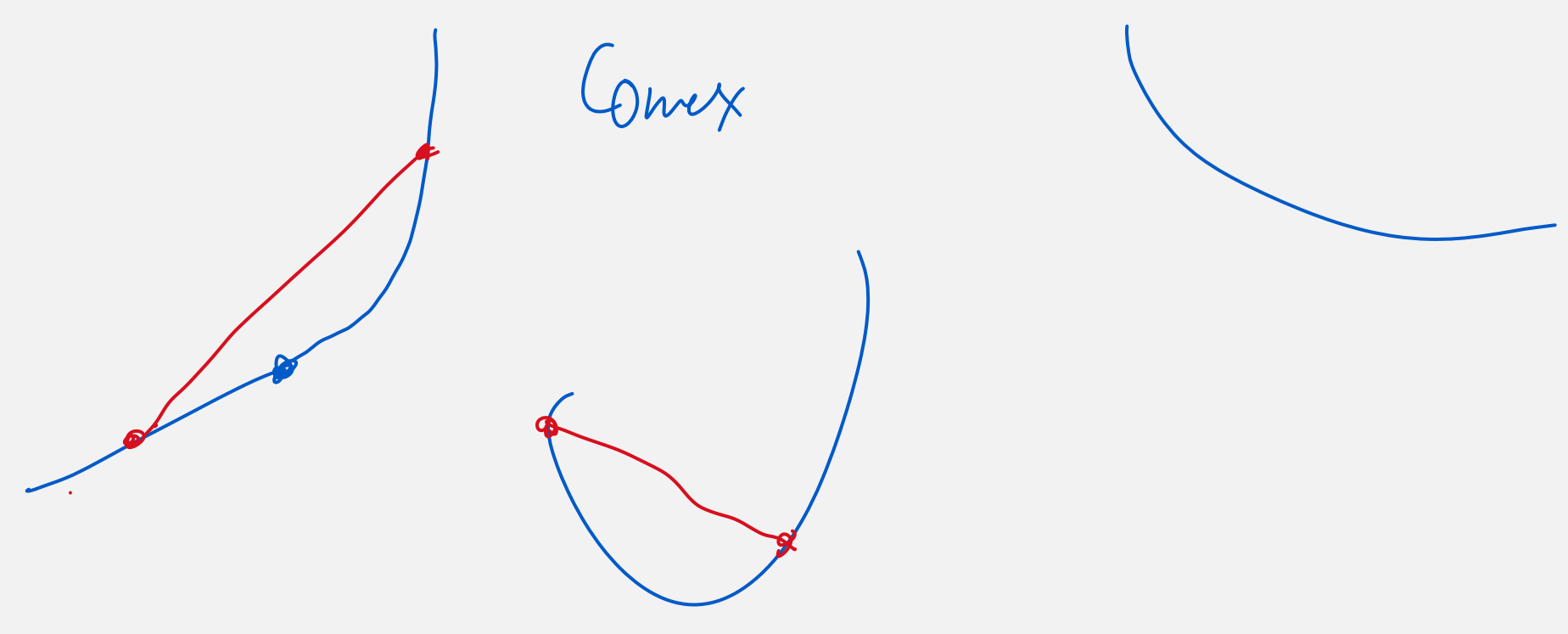
$$B_H(x, x) = (x^T H x) = a \in \mathbb{R}$$ $(1 \times n)(n \times n)(n \times 1)$ → $x^T H x \geq 0. \ \forall x$

$x \cdot y = x^T y = x^T I y$ "dot product"

pos. definite = strict convex: $x^T H x > 0 \quad \forall \vec{x} \neq \vec{0}$

Convex: $f(x) \leq$ line from $f(a)$ to $f(b)$

(strictly: $<$)



Concavity detects convexity.